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Chemistry Paper 3A WASSCE (SC), 2016

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Question 1A

1 A is a solution of $0.050 \text{ mol dm}^{-3} \text{ H}_2\text{C}_2\text{O}_4$

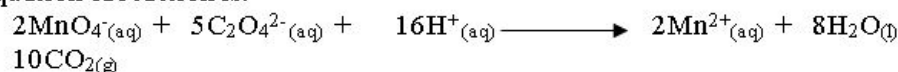
B is a solution of KMnO_4 (potassium tetraoxomanganate (VII)), of unknown concentration.

(a) Put B into the burette. Pipette 20.0 cm^3 or 25.0 cm^3 of A into a conical flask and add about 10.0 cm^3 of dilute H_2SO_4 . Heat the mixture to about 40°C – 50°C and titrate it while still hot with B.

Repeat the titration to obtain consistent titre values.

Tabulate your results and calculate the average volume of B used.

The equation of reaction is:



(b) From your results and the information provided, calculate the:

- concentration of MnO_4^- in B in mol dm^{-3} ;
- concentration of MnO_4^- in B in g dm^{-3} ;
- volume of CO_2 involved at s.t.p. when 25.0 cm^3 of $\text{H}_2\text{C}_2\text{O}_4$ reacted completely.

[O = 16.0, K = 39.0, Mn = 55.0, Molar volume of gas at s.t.p. = $22.4 \text{ dm}^3 \text{ mol}^{-1}$]
[21 marks]

Observation

The question is a redox titration. Majority of candidates were not familiar with the demands of the question, and the performance was fair.

In part (a), majority of the candidates were able to tabulate their burette readings. However, some candidates' readings were not consistent and concordant.

In part (b), some candidates mistook redox reaction for acid-based titration. Hence, they found it difficult to get the correct mole ratio for the reactants, and the

molar concentration for KMnO_4 . Besides, majority of the candidates could not calculate correctly the volume of CO_2 evolved at s.t.p. when 25.0 cm^3 of $\text{H}_2\text{C}_2\text{O}_4$ reacted completely.

Expected answers include:

(a) Two consistent/ concordant titres

Averaging

(b) (i) from the equation of reaction

$$\frac{C(\text{MnO}_4^-) \times V(\text{MnO}_4^-)}{(C_2\text{O}_4^{2-}) \times V(C_2\text{O}_4^{2-})} \text{ OR } \frac{n(\text{MnO}_4^-)}{n(C_2\text{O}_4^{2-})} = \frac{2}{5}$$

$$\begin{aligned} \therefore C(\text{MnO}_4^-) &= \frac{2 \times C(C_2\text{O}_4^{2-}) \times V_{C_2\text{O}_4^{2-}}}{5 \times V_{\text{MnO}_4^-}} \\ &= \frac{2 \times 0.05 \times 25/20}{5 \times V_B} \\ &= a \text{ mol dm}^{-3} \end{aligned}$$

(ii) Molar Mass of $\text{KMnO}_4 = 39 + 55 + (16 \times 4)$

$$= 158.0 \text{ g mol}^{-1}$$

$$C = \frac{\rho}{m} \therefore \rho(\text{KMn}_4\text{O}) = C \times M$$

$$= a \times 158$$

$$= b \text{ g dm}^{-3}$$

(iii) Volume of CO_2 evolved

$$\frac{n(\text{CO}_2)}{n(\text{H}_2\text{C}_2\text{O}_4)} = \frac{10}{5} = \frac{2}{1}$$

$$n(\text{CO}_2) = \frac{2}{1} \times n(\text{H}_2\text{C}_2\text{O}_4)$$

$$= 2 \times 0.0500$$

$$= 0.100 \text{ mol}$$

$$V(\text{CO}_2) \text{ at s. t. p} = n \times V_m$$

$$25 \text{ cm}^3 \text{ of } \text{H}_2\text{C}_2\text{O}_4 \equiv 0.100 \text{ mol} \times 22.4 \text{ dm}^3 \text{ mol}^{-1} \times \frac{25}{1000}$$

$$= 0.056 \text{ dm}^3 \text{ or } 56.0 \text{ cm}^3$$

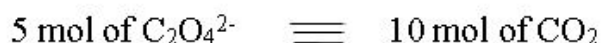
ALTERNATIVE METHOD

Volume of CO_2 evolved

$$\text{Amount of } \text{H}_2\text{C}_2\text{O}_4 \text{ used} = 0.050 \times \frac{25}{1000}$$

$$= 0.00125 \text{ mol}$$

From the equation of the reaction



$$0.00125 \text{ mol } \text{C}_2\text{O}_4^{2-} \equiv 0.00125 \text{ mol} \times \frac{10}{5} = 0.00250 \text{ mol of } \text{CO}_2$$

At S.T.P

$$1 \text{ mol of } \text{CO}_2 \equiv 22.4 \text{ dm}^3$$

$$0.00250 \text{ mol} \equiv 22.4 \times 0.0025$$

$$= 0.056 \text{ dm}^3 \text{ or } 56.0 \text{ cm}^3$$

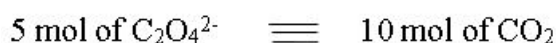
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Volume of CO_2 evolved

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At S.T.P

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$$= 0.056 \text{ dm}^3 \text{ or } 56.0 \text{ cm}^3$$

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